

Vincent Hwang

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🐙 [vincentvbh](https://github.com/vincentvbh)

Profile

Cryptographic engineer, PhD (Max Planck Institute for Security and Privacy, 2026). I optimize post-quantum cryptography at the assembly level on Arm Cortex-M, Armv8-A Neon, AVX2, and CUDA, and formally verify the results with CryptoLine. 16 peer-reviewed papers, 7 at TCHES, plus IEEE S&P and USENIX Security.

Education

- Jan. 2023 – Apr. 2026 **PhD, Cryptographic Engineering**, *Max Planck Institute for Security and Privacy*, Bochum, Germany
Defended April 2026 at Radboud University (external candidate).
Thesis: vincentvbh.github.io/dissertation.pdf.
- Sep. 2021 – Jun. 2022 **MSc, Computer Science and Information Engineering**, *National Taiwan University*, Taipei, Taiwan
- Sep. 2016 – Jun. 2021 **BSc, Computer Science and Information Engineering**, *National Taiwan University*, Taipei, Taiwan

Experience

- Apr. 2026 – present **Senior Software Engineer**, *Qinivicta Inc.*, Boston, USA, Full time
- Jan. 2023 – Mar. 2026 **PhD Student (6 papers)**, *Max Planck Institute for Security and Privacy*, Bochum, Germany, Full time
Formal verification of optimized cryptographic code in EasyCrypt and CryptoLine ($\times 2$), a sole-authored survey of polynomial multiplication for lattice-based cryptography ($\times 1$), and optimized post-quantum implementations for microcontrollers ($\times 1$), high-performance processors ($\times 1$), and portable code ($\times 1$); published at IEEE S&P, USENIX Security, TCHES, CiC, ACISP, and IWSEC.
- Jul. 2022 – Dec. 2022 **Research Assistant (2 papers)**, *Academia Sinica*, Taipei, Taiwan, Full time
Vectorized polynomial multiplication for NTRU and NTRU Prime on Armv8-A Neon (Cortex-A72, Apple M1); published at ACNS and INDOCRYPT.
- Feb. 2022 – Jun. 2022 **Teaching Assistant**, *National Taiwan University*, Taipei, Taiwan, Voluntary
Designed the final project scope and its evaluation.
- Jul. 2021 – Aug. 2021 **Research Assistant (6 papers)**, *Academia Sinica*, Taipei, Taiwan, Full time
NTT-based polynomial multiplication for Kyber, Dilithium, Saber, NTRU, and NTRU Prime on Cortex-M3/M4 ($\times 4$) and on Cortex-A72 and Apple M1 with Neon ($\times 1$), and formally verified NTTs with CryptoLine ($\times 1$); published at TCHES ($\times 4$), ACNS, and IWSEC.
- Jul. 2020 – Aug. 2020 **Summer Intern (2 papers)**, *Academia Sinica*, Taipei, Taiwan, Full time
NTT-based polynomial multiplication for NTRU Prime, NTRU, and Saber on Cortex-M4, with speed records for Saber and NTRU; published at TCHES.

Skills

Programming Assembly (Armv7-M, Armv7E-M, Armv8-A, Armv9-A, AVX2, AVX-512), C/C++, CUDA, Jasmin, Haskell, Sage, Python 3.
Tools Arm GNU toolchain, GCC, Clang, GNU Make, GNU Bash, CryptoLine.

Publications

2021 – 2026 Full list of publications (16 papers): vincentvbh.github.io/pub.pdf.

Academic Service

Editorial board

2027 TCHES 2027, Program Committee / Editorial Board Member.

Reviewer

2026 TCHES 2026 ($\times 8$), Eurocrypt 2026 ($\times 1$), Crypto 2026 ($\times 2$).

2025 TCHES 2025 ($\times 8$), ArcticCrypt 2025 ($\times 1$), CT-RSA 2025 ($\times 2$), JCEN ($\times 1$).

2024 TCHES 2024 ($\times 3$), Crypto 2024 ($\times 1$).

Artifact evaluation

2026 TCHES 2026 ($\times 6$), Eurocrypt 2026 ($\times 4$).

2025 TCHES 2025 ($\times 5$).

2023 TCHES 2023 ($\times 2$).

TCHES = IACR Transactions on Cryptographic Hardware and Embedded Systems; CT-RSA = Cryptographers' Track at the RSA Conference; JCEN = Journal of Cryptographic Engineering; ($\times n$) = number of reviews submitted.

Languages

Mandarin Native.

English Full professional proficiency.